GREEN CITY HACKATHON · PROJECT

Sag's Heilbronn

Citizens report where Heilbronn is too loud, too hot or the air is bad. We check it against sensor data. Where no data exists, a sensor goes there.

> WHO: **The Green Flags**
> FOR WHO: **Heilbronn citizens**
> WHAT: **Report · Cluster · Check**

THE STORY IN THREE NUMBERS

40,200

Heilbronn residents are exposed to road traffic noise above 55 dB(A) Lden.

Lärmaktionsplan Stufe IV, 2024

Ungeprüft

"Unchecked": the status of noise complaints about racing and "Poser" nights in the city's Mängelmelder (defect reporting tool), filed under "Straße und Verkehr". Nobody can check them.

Mängelmelder Heilbronn

2029 → 2026

The next comparable noise action plan evaluation is due in 2029 – by calculation, not measurement. We start evaluating now, in 2026, by measuring.

Lärmaktionsplan Stufe IV

Scan, report, done.

greencity.parsalotfi.com

Three taps, location from GPS. It is live now – try it right here at the table.

How it works

01 · MELDEN

Report

Three taps, location from GPS. What bothers you here?

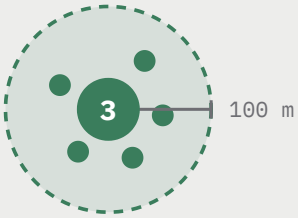
- Too loud
- Too hot
- Bad air
- No shade
- Litter*
- Broken road*
- Other

* Outlook: the same loop works for anything a city can measure.

02 · BÜNDELN

Cluster

Reports within 100 m form one spot. The number on the map is the number of reports.



03 · PRÜFEN

Check

Each spot is compared with the nearest sensor; shade against the city's tree register.

- **Bestätigt**
- **Nicht bestätigt**
- **Sensor fehlt**

Where the sensor is missing, reports decide: **"Platz 1 von 8 Stellen ohne Sensor"** (no. 1 of 8 spots without a sensor) – that is where the next sensor goes.

WHAT EXISTS TODAY

- **The app**, live on a Raspberry Pi in Heilbronn.
- **Data** from openSenseMap (senseBox:home stations and senseBox:bike rides) and the Umweltbundesamt station DEBW015.
- The city's **noise map 2022** and **tree register** as map layers.
- **Our own sensor "NoiseNode"**: senseBox MCU-S2 + sound level meter, 3D-printed case, about 100–150 € in parts. Live on openSenseMap since 28 October 2026, measuring LAeq, Lmax and loud events per minute. No audio is recorded.
- **Open source on GitHub:** github.com/HelloX3030/green-city

THE ASK

The next **10 NoiseNodes** for the streets of the Lärmaktionsplan – measured before and after the planned low-noise asphalt (2.5 million € in the plan).

So the city has **a measured answer in 2027** instead of a recalculation in 2029.

{ A greener city isn't just one that collects more data. It's one that listens to its people and acts on what it learns. }

TEAM	The Green Flags: Ethan, Alexia, Lukas, Fred, Seyedparsa
HACKATHON	Green City Hackathon Heilbronn, 27–29 October 2026
DATA	openSenseMap, Umweltbundesamt, Stadt Heilbronn
CODE	Open source on GitHub · github.com/HelloX3030/green-city